

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 214 — Past Questions & Materials
by Renance

Course: MTH 214
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 6
Levels: 200, 300

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Paper 1 of 6 — MTH 214 Statistics for Physical Science and Engineering First Semester 2020 2021

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS / COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2020/2021
Course Code: MTH 214 Title: Statistics for Physical Science and Engineering
Level: 200

Instruction: Answer any four questions.

Duration: $2\frac{1}{2}$ hrs

1. (a) Explain the following statistical concepts: (4 Marks)
(i) Factor (ii) Treatment (iii) Experimental Unit (iv) Pairing

- (b) An automobile manufacturer is comparing the failure times (in hours) of two different makes of alternators. The data are listed below for five units of each type. Test the significant difference of the mean time to failure of brakes from the two manufacturers at $\alpha = 0.01$. (5 Marks)

Make	Time to Failure (hrs)				
Manufacturer A	131	127	110	76	40
Manufacturer B	65	51	18	12	5

- (c) Test the significant difference of the two sample means using Mann-Whitney test. (8.5 Marks)

2. (a) The data in the table below show the number of bedrooms, the number of births and two years charge for renting a flat in one particular city in Nigeria.

No. of bedrooms (X1)	3	2	4	2	3	2	5	4
No. of baths (X2)	2	1	3	1	2	2	3	2
Price (Y, in thousand)	788	743	838	742	797	749	884	829

- Fit a linear model of the form $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$ for the data given that (7 Marks)

$$(X'X)^{-1} = \begin{pmatrix} 1.2738 & -0.2381 & -0.2024 \\ -0.2381 & 0.3810 & -0.4762 \\ -0.2024 & -0.4762 & 0.8452 \end{pmatrix}$$

- (b) Test the significant of the overall model given that $\alpha = 0.05$. (10.5 Marks)

3. Three different washing solutions are been compared to study their effectiveness in retarding bacteria growth in five litres milk containers. The day the experiment is performed could represent a potential source of variability. The order in which the solution were tested on a particular day were randomly determined. The data obtained is presented in the table below. Using $\alpha = 0.05$.

Washing Solution	Days			
	1	2	3	4
1	13	22	18	39
2	16	24	17	44
3	5	4	1	22

- (a) Test the hypothesis that there is no significant difference in efficiency of the three solutions: (11.5 Marks)

- (b) Test the hypothesis that there is no significant difference in output of the four days. (2.5 Marks)

- (c) If the null hypothesis is rejected in (a), determine the pair of solutions that is responsible for the rejection. (3.5 Marks)

4. The Yield of a chemical process is being studied. The two most important variables are thought to be the pressure and temperature. Three levels of each factor are selected, and a factorial experiment with two replicates performed. The yield data are presented in the table below:

Paper 2 of 6 — mth 214 past question

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS / COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2019/2020

Course Code: MTH 214 Title: Statistics for Physical Sciences and Engineering
Level: 200

Instruction: Answer any four questions.

Duration: $2\frac{1}{2}$ hrs

1. (a) Explain the following statistical concepts:

- (i) Statistical survey (ii) Sample statistic (iii) Population parameter (3 Marks)
(b) In an experiment, we wish to find the sample size that is needed to make the error bound in a 95 percent confidence interval for mean equal to 0.3. The standard deviation of preliminary sample of size 35 is 0.7517. (5 Marks)
(c) An automobile insurance company wishes to compare the repair costs of moderately damaged cars. A random sample of size 7 were selected and taken to garage 1 and then to garage 2 for cost (thousands naira) estimation and presented in the table below. (9.5 Marks)

Garage 1	7.1	9.0	11.0	8.9	9.9	9.1	10.3
Garage 2	7.9	10.1	12.2	8.8	10.4	9.8	11.7

Test at $\alpha = 0.05$ whether the two garage have equivalent repair costs.

2. (a) List the advantage of replication and randomization in a designed experiment. (4 Marks)
(b) A marketing research firm wishes to compare the prices charged by two supermarket chains. Identical purchase were made from 10 of each chain's stores. The stores for each chain are randomly selected and all purchases were made during a single week. The shopping expenses (thousands) are as presented below. Test the significant difference of the mean shopping prices from the two supermarket chains at $\alpha = 0.01$. (6.5 Marks)
Chain 1: 119.25, 121.32, 122.34, 120.14, 122.19, 123.71, 121.72, 122.42, 123.63, 122.44.
Chain 2: 111.99, 114.88, 115.11, 117.02, 116.89, 116.62, 115.38, 114.40, 113.91, 111.87.
(c) Test the significant difference of the two mean shopping prices using Mann-Whitney test. (7 Marks)

3. (a) An oil company wishes to compare the effects of three types of unleaded gasoline (A, B and C) on gasoline mileage. Each randomly selected car was test driven using the appropriate gasoline type under normal conditions for specified distance, and the gasoline mileage measured. Test the significant of the gasoline types on mileage at $\alpha = 0.05$. (9 Marks)

A	34.0	35.0	34.3	35.5	35.8
B	35.3	36.5	36.4	37.0	37.6
C	33.3	34.0	34.7	33.0	34.9

- (b) Test the significant difference of the three sample means using Kruskal Wallis test. (8.5 Marks)

4. (a) A real estate agency collects the following data concerning sales price of a house, in hundred thousands of naira (y), home size, in hundreds of square feet (x_1) and rating (x_2) expressed on a scale from 1 (worst) to 10 (best).

X_1	23	11	20	17	15	21	24	13	19	25
X_2	5	2	9	3	8	4	7	6	7	2
Y	120	65.4	115.4	91	94	110.6	129	85.2	109	115

Fit a linear model of the form $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$ for the data given that

$$(X'X)^{-1} = \begin{pmatrix} 2.276908 & -0.091269 & -0.086990 \\ -0.091269 & 0.004970 & -0.000407 \\ -0.086990 & -0.000407 & 0.017859 \end{pmatrix}$$

- (b) Test the significant of the home size in the model given that $\alpha = 0.05$. (9 Marks)

5. A consumer preference study involving three different bottle designs (A, B and C) for the jumbo size of a new liquid laundry detergent was carried out in four supermarkets. The results of the quantity sold for each design and supermarket are presented in the table below and analysis is carried out at $\alpha = 0.05$.

Bottle Design	Supermarket			
	1	2	3	4
A	16	14	1	6
B	33	30	19	23
C	23	21	8	12

- (a) Test for the significant difference in quantity sold for the three bottle designs. (11 Marks)
 (b) Test for the significant difference in quantity sold in the four supermarkets. (2 Marks)
 (c) If the null hypothesis is rejected in (a), determine the pair of bottle designs that is responsible for the rejection using Least Significant Difference (LSD) method. (4.5 Marks)

6. An experiment is performed to determine the effects of four brands of baking flours and three brands of yeasts on the weight of baked bread. Four replications were made for each flour and yeast combinations. The data are presented in the table below.

Baking Flour	Yeast					
	A		B		C	
1	13.7	17.9	39.3	30.5	28.3	29.5
	17.4	17.9	28.6	27.6	29.3	28.4
2	26.8	25.5	29.9	30.4	30.7	29.7
	26.1	25.7	28.6	29.1	30.4	30.4
3	24.9	25.7	27.9	27.5	28.5	26.5
	25.8	26.1	27.6	26.8	28.2	26.9
4	22.6	23.7	31.4	32.0	31.8	31.1
	22.6	23.0	30.8	32.8	30.0	30.2

Using $\alpha = 0.05$.

- (a) Test the null hypothesis that there is no significant difference among the four levels of baking flour. (13.5 Marks)
 (b) Test the null hypothesis that there is no significant difference among the three levels of yeast. (2 Marks)
 (c) Test the hypothesis about the significant of interaction between baking flour and yeast. (2 Marks)

Best of Luck.

Paper 3 of 6 — MTH 214 ASSIGNMENT

Level 300 • First Semester • PastQuestion

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Chemical	8015				
	1	2	3	4	5
1	78	68	74	71	67
2	73	67	75	72	70
3	75	68	78	73	68
4	73	71	75	75	69

2. An industrial engineer is conducting an experiment on eye focus time. He is interested in the effect of the distance of the object from the eye on the focus time. Four different distances are of interest. He has five ^{individuals} subjects available for the experiment. Because there may be differences between individuals, he decides to conduct the experiment in a randomized block design. The data obtained are shown below. Analyze the data and draw appropriate conclusions using $\alpha = 5\%$.

Distance (m)	Individuals				
	1	2	3	4	5
4	10	6	6	6	6
6	7	6	6	1	6
8	5	3	3	2	5
10	6	4	4	2	3

	SS	df	MS
Between Treatments	412.64867	3	137.5482
Within Treatments (Error)	543.097	27	20.1147
Total	955.7457	30	

$$F_{\text{tab}} \Rightarrow F_{0.05, 3, 27} = 2.92$$

Decision: ~~we cannot~~ Since $F_{\text{cal}} > F_{\text{tab}}$, the null hypothesis must be rejected. We conclude that the performances of the four typewriters differ significantly and that the difference is not due to chance.

$$3. = \frac{2}{1} \times \frac{4}{1} \times \frac{5}{1}$$

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- ① Tutorial 2
The following are the numbers of mistakes made in five successive weeks by four technicians working for a medical laboratory:

Technician I: 13, 16, 12, 14, 15

✓ II: 14, 16, 11, 19, 15

✓ III: 13, 18, 16, 14, 18

✓ IV: 18, 10, 14, 15, 12

Test at the 0.05 level of significance whether the differences among the four sample means can be attributed to chance.

- ② A consumer testing service intends to test the accuracy of the thermostats of three different kinds of electric iron; set them at 480°F and obtained the following actual temperature readings by means of a thermocouple:

Iron X: 474, 496, 467, 471

Iron Y: 492, 498

Iron Z: 460, 495, 490

Test at the 0.05 level of significance whether the differences among the three sample means can be attributed to chance.

Paper 4 of 6 — MTH 214 2022/2023

Level 200 • First Semester • PastQuestion

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FEDERAL COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS

END OF SEMESTER EXAMINATION FOR MTH 214- (Sta. for Physical Sci. and Engr.) 2022/2023

Instruction: Answer any 4 questions. Date of Examination: 24/05/2023 Times: 2 hours 15 minutes

- (a) Let $F(x) = 1 - \exp\left(-\frac{\theta}{\beta}(1 - \exp(\beta x))\right)$ be the probability density function (pdf) and cumulative distribution function (cdf) for a random variable x of the Gompertz distribution. Compute the quantile function of the Gompertz distribution. (10 marks)

(b) Given the exponential probability density function (pdf) $f(x) = \lambda \exp(-\lambda x)$ and a cumulative distribution function (cdf) $F(x) = 1 - \exp(-\lambda x)$ for a random variable x . Obtain the quantile function of the exponential distribution. (7.5 marks)
- (a) Let exponential probability density function (pdf) $f(x) = \lambda \exp(-\lambda x)$ and a cumulative distribution function (cdf) $F(x) = 1 - \exp(-\lambda x)$ for a random variable x . Obtain the Order statistic and the Hazard rate function.

(b) Suppose the Weibull probability density function (pdf) is given $f(x) = \alpha \beta x^{\beta-1} \exp(-\alpha x^\beta)$ and a cumulative distribution function (cdf) $F(x) = 1 - \exp(-\alpha x^\beta)$ for a random variable x . Compute the reversed hazard rate function.
- Let exponential probability density function (pdf) $f(x) = \lambda \exp(-\lambda x)$ and a cumulative distribution function (cdf) $F(x) = 1 - \exp(-\lambda x)$ for a random variable x . Also, let $g(x) = \alpha \beta f(x) \frac{[F(x)]^{\beta-1}}{[1-F(x)]^{\beta+1}} \exp\left\{-\alpha \left[\frac{F(x)}{1-F(x)}\right]^\beta\right\}$ and $G(x) = 1 - \exp\left\{-\alpha \left[\frac{F(x)}{1-F(x)}\right]^\beta\right\}$ be the Weibull-G pdf and cdf. Obtain the Weibull exponential distribution pdf and cdf.
- (a) the length of the Udu bridge is assumed to follow a continuous random variable with a probability density function (pdf) $f(x) = \frac{3x^2}{2} + x$, $0 \leq x \leq 1$. Show that $f(x)$ is a pdf.

(b) Find the moment generating function for the continuous random variable for the pdf $f(x) = 4 \exp(-4x)$, $0 \leq x \leq \infty$.
- (a) The daily expenses of a company has a probability density function which is given by $f(x) = \frac{3}{26}(1+x)^2$, $0 < x < 2$. What is the expected amount to be budgeted for in 5 days in thousands of naira.

(b) Five maize seeds are planted in a row in which the first, third and fifth seeds germinated and the other two did not. If the probability of germination is 0.8. What is the probability of the outcome assuming independence?

A	B	C	D
19	14	11	24
21	16	14	19
26	14	21	21
24	13	13	26
18	17	16	20
	13	18	

At a 5% significance level, test the null hypothesis that the mean number of customers served per hour by each of these four tellers is the same. Assume that all the assumptions required to apply the one-way ANOVA procedure hold true.

Paper 5 of 6 — Mth 214 by Kess.

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Paper 6 of 6 — MTH 214

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